

Realistic Synthetic Data Generation for Defective Radiographic Testing Scans of Welds

Master's Thesis Topic Proposal — v0.2, September 2026

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Motivation

Radiographic testing (RT) is the standard non-destructive method for inspecting welded joints in pipelines and pressure vessels. Deeplify GmbH automates its evaluation with segmentation models that localise six defect classes on digitised radiographs. The dominant bottleneck is not architecture but **data**: annotated defective film is scarce, expensive and extremely imbalanced.

The public SWRD corpus [1] used throughout this thesis makes the problem concrete. Across its 4,930 annotated weld images, porosity is annotated on 3,359 films while undercut appears on only 132; at instance level the ratio is 26,674 to 228. A single defect polygon covers a median of 0.024% of a film. A production SegFormer baseline trained on this corpus reaches a macro IoU of 0.484, and its per-class spread exposes **two distinct failure modes** rather than one:

class	films	instances	median area	baseline IoU
lack of penetration	483	796	0.57%	0.730
crack	1,237	2,071	0.13%	0.742
porosity	3,359	26,674	0.019%	0.621
lack of fusion	418	857	0.19%	0.415
undercut	132	228	0.12%	0.296
inclusion	1,019	2,202	0.041%	0.239

Scarcity explains undercut and lack of fusion: too few examples of a subtle, low-contrast indication. **Ambiguity** explains inclusion, which is common yet scores worst — the same corpus carries 13,311 *pseudo-defect* annotations (film artefacts that mimic defects), and the model cannot separate the two. Lack of penetration is rare but easy, because it is large and stereotyped. A controllable generator can attack both modes at once: mint rare defects on demand, and mint the confusable artefacts that teach the model where the boundary lies.

Thesis Goal

Develop and evaluate generative methods that synthesise realistic defective RT scans — image plus pixel-accurate defect mask — such that training on real + synthetic data measurably improves downstream defect segmentation, especially for the rare and the confusable classes. The scientifically distinctive element is a **physics-informed formulation**. A radiograph follows Beer–Lambert attenuation ($I = I_0 \cdot \exp(-\int \mu \, dl)$), so in log-intensity space a real defect is an *additive, sign-constrained* perturbation set by its thickness profile, not an arbitrary texture.

The gap must be stated precisely, because physically grounded compositing into real transmission images is not new. Security screening derives it from Beer–Lambert explicitly: threats are projected into real cargo radiographs by *multiplication* in transmission [11], equivalently by *addition of logarithmic images* [12], with augmentations that are themselves power laws on transmission. NDT reached the same idea first, superimposing CAD models of flaws onto real castings and weld radiographs [13]. Medicine inserts simulated lesions into real mammograms and CT projection data and validates them against readers and ground-truth morphometry [14, 15, 16].

What no one has done is the part this thesis proposes. **First**, every physically grounded pipeline above composites an *externally specified* object — a CAD flaw, a scanned threat — and none learns the defect appearance distribution; conversely every learned defect generator composites without any attenuation constraint. **Second**, and most testable, the decisive experiment is missing: no published work trains one model on physically composited synthetic data and an otherwise identical model on naively blended synthetic data and compares both on real images. The one study that used naive alpha blending lost 10 mAP against real

training data, found that mixing real and synthetic made things worse, and closed by suspecting its own compositing quality without ever testing the physical alternative [17]. **A physically constrained, mask-emitting generator for weld radiography, with that A/B as its core ablation, is the contribution.**

Data, and an Honest Constraint

All experiments use **SWRD only** [1] — 3,675 scanned films of longitudinal and T-joint seam welds, cropped to 4,930 annotated weld images, 16-bit. This keeps every result reproducible outside Deeplify and avoids conditioning the thesis on annotation quality we are still auditing internally. SWRD is unusually well suited to this work: **every film ships a weld-seam polygon**, so defect placement priors come free with no auxiliary model; 353 weld images are annotated defect-free and the remaining films contain long defect-free bead sections, giving abundant authentic canvases for injection; and the pseudo-defect class supplies a ready-made hard-negative target.

One constraint must be stated up front, because it shapes the physics-informed work. **SWRD ships no acquisition metadata.** Verified against both the dataset paper and the files themselves: the paper reports only the technique (vertical single-wall radiography of large pipeline seams, MIG welds) and the film scanner (Microtek MII5000LC, industrial CCD, 1200 dpi, 16 bit); the TIFFs carry nothing but geometry and a default 96 dpi written by a Windows image viewer; the LabelMe JSON files carry polygons and nothing else. Material grade, wall thickness, pipe diameter, tube voltage, exposure and source-to-film distance are absent. Spot-checking industrial DICOM files does not rescue this — the header fields exist there but are left unfilled by the operator (tube voltage present in 1 of 60 sampled files, source-to-detector distance in none, the DICOM component-thickness attributes in none); only pixel spacing is reliably recorded.

The response is to make the formulation **metadata-free by construction**. The physical constraints worth enforcing — additivity in log space, a fixed perturbation sign, a smooth thickness profile, correct photon and grain noise — do not require μ , t or geometry to be known. Only the *scale* does, and the scale can be calibrated from the film itself: the weld cap is a step of approximately known physical height that welding codes bound, and the uncropped originals carry wire-type image quality indicators and lead markers of standardised dimensions. Recovering an attenuation scale this way, and testing whether a generator constrained by it beats one that is not, is itself a contribution.

State of the Art (to be surveyed in depth)

Physics-based simulators (aRTist [2], gVirtualXray) give exact physics and free labels but suffer a sim-to-real appearance gap and need explicit defect geometry — though SyntheX [18] shows that with enough domain randomisation, models trained purely on simulated X-rays can beat models trained on real ones. **GAN-based defect synthesis** (Defect-GAN [3], few-shot DFMGAN [4]) is fast but unstable in exactly the low-data regime that matters here. **Diffusion models** now dominate controllable synthesis: for industrial anomalies AnomalyDiffusion [5] and RealNet [6] generate anomaly–mask pairs, while mask-conditioned **inpainting** [7, 8] injects defects into abundant real defect-free film and inherits authentic grain, scanner response and weld texture — the approach this thesis treats as its primary learned method. Two 2026 journal papers now apply conditional diffusion to weld and surface defect generation [19, 20] and define the competitive bar. **Medical imaging** has attacked the identical problem of tiny, rare, expensive lesions for longer than industrial inspection has — lesion-focused diffusion that constrains generation to the given mask [21], hand-crafted synthesis that rivals real training data [22], and the insertion-plus-validation tradition above — and is surveyed as a first-class source of technique rather than as related work. None of these enforce radiographic attenuation structure on a learned generator, which is where the proposed hybrid [9, 10] sits.

Research Questions

RQ1 Which paradigm — physics simulation, GAN, or mask-conditioned diffusion inpainting — yields the most *useful* synthetic weld defects under one common evaluation?

RQ2 Can class- and mask-conditioned inpainting inject convincing defects into real defect-free SWRD film, using the corpus's own seam polygons as the placement prior, with a mask that stays exact?

RQ3 Do physics-informed constraints — attenuation consistency in log space, a sign constraint, generation in a thickness-perturbation space, photon and grain noise realism — improve realism and utility, given that the attenuation scale must be calibrated from the image rather than read from metadata? Ablations isolate each term, and the headline comparison is the missing A/B above: identical training, physically composited synthetic

data versus naively blended, scored on real film.

RQ4 Does it help downstream? Primary metric: per-class IoU and instance detection rate on a fixed, purely real SWRD validation set, against the existing real-only baseline (macro IoU 0.484), swept over several synthetic fractions. Secondary: FID/KID and a blinded realism study with certified RT inspectors.

Setting and Work Plan

Deeplify provides a production training and evaluation stack with an established SWRD baseline, GPU infrastructure, and access to certified inspectors, so results plug into a fixed honest benchmark rather than a synthetic one. Preliminary 6-month plan:

M1–M2	Survey; evaluation protocol; real-only and classical-augmentation baselines; diffusion inpainting pipeline on defect-free SWRD fil
M3–M4	Physics-informed formulation, attenuation-scale calibration and ablations; scale generation; retrain and measure utility
M5–M6	Expert realism study; rare-class and hard-negative experiments; thesis writing

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SWRD class statistics were measured directly on the corpus's 4,930 annotation files; baseline IoU figures are Deeplify-internal measurements on SWRD. Scope and methodology remain open for discussion.